

Microwave radiation has modulation frequency dependent stimulating effect on human EEG rhythms

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Abstract— This study is focused on low-level modulated microwave field effects on human EEG theta, alpha and beta rhythms at different modulation frequencies. During the experiment 13 healthy volunteers were exposed to a microwave (450 MHz) with 7 Hz, 14 Hz and 21 Hz frequency on-off modulation. The field power density at the scalp was 0.16 mW/cm². The experimental protocol consisted of five cycles of the repetitive microwave stimulation at fixed modulation frequencies. Changes in the EEG rhythms energy became evident in the case of modulation frequencies higher than the EEG rhythms frequencies. The changes varied strongly from subject to subject. Microwave exposure caused statistically significant changes in the EEG theta rhythm energy and for occipital channels in the alpha rhythm energy.

Keywords— Electromagnetic field, EMF effects, non-ionizing radiation, EEG rhythms, alpha, theta.

I. INTRODUCTION

The wide use of microwave devices (mobile phones, microwave cooking etc) by population as well as application of electric and transcranial magnetic stimulation in brain research and neurotherapy has raised an interest to the possible effects of electromagnetic fields (EMF). During recent years, non-thermal effect of low-level EMF on human nervous system has become a subject of discussions. The reports of possible non-thermal EMF effects are often contradictory. Several investigators have reported that low-level exposure produces alterations in the EEG signal and brain behavior [1-8]. The conclusion reported by other researchers is that the exposure to EMF does not alter the resting EEG [9,10, 11].

Mechanisms behind the effects are still unclear. Changes in brain behavior state are typically accompanied by changes in the frequency and spatial coordination of rhythmic activity in the neocortex and the efficacy of signal transmission through electrical coupling has been found to be frequency dependent [12, 13]. The hypothesis raised in this study was following: if the EMF as a weak stressor causes any effect on human brain electrical activity, that effect should depend on stimulation frequency.

The radiation of cellular phones is modulated inside a wide frequency band. Differences of the effects caused at separate modulation frequencies cannot be distinguished in the case of experiments with cellular phones. During special experimental studies with other EMF sources usually only one modulation frequency has been used. Unfortunately the results achieved at different modulation frequencies in

different experimental studies are not comparable due to the diversity of experimental conditions.

The aim of this study is the evaluation of the microwave field effect on human EEG rhythms at different modulation frequencies. According to the hypothesis, the effect, if related to the changes in neocortical dynamics, should depend on modulation frequency. Modulation frequencies of 7 Hz, 14 Hz and 21 Hz were selected and EEG theta, alpha and beta rhythms analyzed.

II. METHODOLOGY

A. Subjects

The experiments were carried out on a group of healthy volunteers. The group consisted of 13 young (age 21-30 years) persons – 4 male and 9 female. A questionnaire and a clinical interview were used to evaluate their physical and mental condition (tiredness, sleepiness) before the experiment. All the subjects selected were without any medical or psychiatric disorders and tired or sleepy persons were excluded. After the recordings, they were asked how they were feeling during the experiment. The subjects reported no alertness or experienced strain during the experiment. The experiments were conducted with the understanding and written consent of each participant.

The experimenter and the subjects were in the same laboratory room during the experiments. The room was dark and the subjects were lying in a relaxed position, eyes closed and ears blocked during the experiments.

All subjects passed the first experimental protocol (with exposure) twice and the second (sham) once. During each test session, the succession of modulation frequencies was randomly assigned. Only one experimental EEG recording was performed for a subject during a day during the time interval from 9 a.m. to noon. The subjects were not informed of their exposure, however, they were aware of the possibility of being exposed.

The study was conducted in accordance with the Declaration of Helsinki and has formally approved by the local Medical Research Ethics Committee.

B. Microwave exposure

Microwave exposure conditions were the same for all subjects and identical to microwave exposure in our previous studies [8, 14].

The 450 MHz microwave radiation was generated by the Rhode & Swartz (Germany) signal generator model SML02. The RF signal was 100% amplitude modulated by

the pulse modulator SML-B3 at 7 Hz frequency (duty cycle 50%). The generator signal was amplified with the Dage Corporation (USA) power amplifier model MSD-2597601. The generator and amplifier were carefully shielded and located in the laboratory room. The 1W EMF output power was guided by a coaxial to the 13cm quarter-wave antenna NMT450 RA3206 by Allgon Mobile Communication AB, Sweden, located at 10 cm from skin from the left side of the head.

The calculation of the specific absorption rate (SAR) inside the brain was based on the known field power density on skin. The spatial distribution of the microwave power density was measured by the Fieldmeter C.A 43 Chauvin Arnoux (France) field strength meter. The calibration curves of the field power density dependence on the distance from the radiating antenna were obtained from these measurements performed in the real condition of the experiment. During the experiments, the stability of the microwave level was monitored by the IC Engineering (USA) Digi Field C field strength meter. Estimated by the measured calibration curves, the field power density at the skin was 0.16 mW/cm². The level of power density as well as the calculated SAR 0.35 W/kg [8] was so low that thermal effects were extremely unlikely.

C. Recording protocols and equipment

The study includes two experimental protocols identical for all subjects.

The first protocol was recorded as follows:

Firstly, the reference EEG was recorded during 10 minutes.

Secondly, the microwave radiation, modulated at first modulation frequency, was applied. The duration of the stimulation was 1 minute and the compensatory pause after the stimulation was also 1 minute. Continuous recordings of the EEG were performed during as well as 1 minute after the stimulation. The procedure of the cycle was repeated five times. The microwave exposure was switched on every first minute of the cycle. During five cycles of the microwave stimulation, the modulation frequency was always the same.

Thirdly, the microwave radiation, modulated at second modulation frequency, was applied. The procedure of this step was identical to the previous one, except the modulation frequency was different. Fourthly, the microwave radiation, modulated at the third modulation frequency, was applied. The procedure of this step was identical to the previous one, except the modulation frequency was different.

The selection of the 7 Hz, 14 Hz and 21 Hz as first, second or third modulation frequency was randomly assigned.

The recording protocol for one subject lasted for 40 minutes, during which the EEG was continuously recorded.

The second protocol for the sham exposure included the same steps, except the microwave generator was switched off.

The Cadwell Easy II EEG measurement equipment was used for the EEG recordings. The EEG was recorded using 9 electrodes, which were placed on the subject's head according to the international 10-20-electrode position classification system. The channels were chosen to cover the entire head: frontal – FP1, FP2; temporal – T3, T4; parietal – P5, P4, occipital – O1, O2 and the reference electrode Cz. The EEG recordings were stored on a computer with a 400 Hz sampling frequency. The recorded EEG signals were observed by an experienced neurologist. The artefacts were detected by visual inspection. The recordings containing multiple artifacts were removed and the recording was repeated. Afterwards, off-line data processing was executed. The energies of the five basic EEG rhythm frequencies (delta, theta, alpha, beta and beta2) were extracted from the total EEG signal (0.5 - 48 Hz) by filtering.

D. Data analysis

The dynamics of the EEG energies of the theta, alpha and beta rhythm frequencies (4-7 Hz, 8-13 Hz, 15-20Hz and 22-40 Hz) were analysed. As one of the modulating frequencies belongs to the same region (7 Hz), it was removed by a narrow band (0.2 Hz) filter. Average energies for segments with and without stimulation were compared. The last 20 seconds of the recording segments with and without stimulation were selected as intervals for comparison since by visual evaluation the most noticeable changes occurred in the last part of the segments. The average energy of an arbitrary comparison interval was calculated for the recording segment as follows:

$$s_i = \frac{1}{N} \sum_{r=1}^N [x(r)]^2, \quad (1)$$

where x is the amplitude of the recorded signal and N is the number of samples, during 20 seconds N=8000. The relative change in the energy of the recording segments with and without stimulation was calculated for further analysis. The relative change was calculated as follows:

$$S = (s_1/s_2 - 1) * 100\%, \quad (2)$$

where s1 and s2 were the average energies inside the comparison intervals with and without stimulation respectively.

The average values of the parameter S for 5 cycles of recordings during two sessions (5+5 minutes) for different EEG channels, for the individuals and for the whole group were calculated.

The two-tailed Student t-test was used to compare the mean values of the differences between stimulated and reference recordings (parameter S) as well as between the segments with and without stimulation within one recording for a subject (values of s1 and s2). All p values lower than 0.05 were considered as significant.

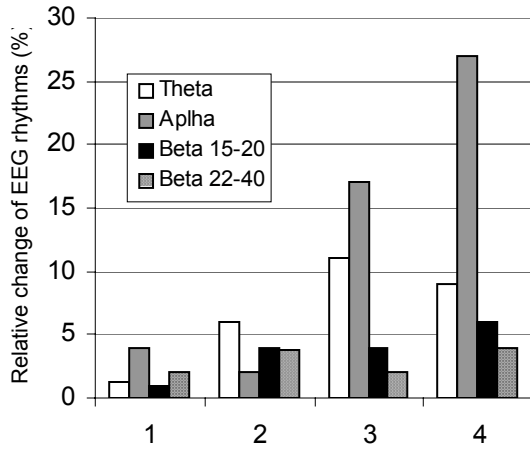


Fig.1 The average value of the relative changes in the EEG rhythms total energy of the recording segments with and without microwave stimulation in 8 EEG channels at different modulation frequencies for a whole group: 1 – reference, 2 – 7 Hz modulation frequency, 3 – 14 Hz modulation frequency, 4 – 21 Hz modulation frequency.

Signal processing and calculation of parameters were performed in the LabVIEW programming and signal processing environment.

III. RESULTS

The average value of the changes caused by microwave stimulation in total energy of the EEG rhythms at different modulation frequencies is presented in Fig. 1. According to the data noticeable changes occurred mainly in the EEG theta and alpha rhythms. In the case of sham recordings all calculated average values were at the level of reference EEG.

Differences between the stimulated and reference recordings were evaluated statistically. The calculated p-values for the averaged values of the parameter S for the whole group are presented in Table 1. The p-values lower than 0.05 regarded as significant are marked bold.

The t-test analysis was also performed for the evaluation of differences between the segments with and without the stimulation for individuals. In the case of eight subjects the p value was less than 0.05 at least in two channels.

TABLE I

CALCULATED P VALUES FOR DIFFERENCES OF AVERAGE VALUES OF THE RELATIVE CHANGES IN THE EEG ENERGY LEVEL BETWEEN REFERENCE AND STIMULATED RECORDINGS FOR THE WHOLE GROUP

Mod. freq.	7 Hz				14 Hz				21 Hz			
EEG freq. Hz	4-7	8-13	15-20	22-40	4-7	8-13	15-20	22-40	4-7	8-13	15-20	22-40
FP1, FP2	0.044	0.513	0.250	0.339	0.026	0.173	0.906	0.992	0.016	0.159	0.754	0.611
TP3, T4	0.185	0.118	0.481	0.422	0.042	0.448	0.876	0.765	0.038	0.121	0.596	0.222
P3, P4	0.043	0.943	0.135	0.481	0.009	0.035	0.249	0.668	0.003	0.048	0.085	0.634
O1, O2	0.175	0.863	0.113	0.466	0.113	0.015	0.216	0.987	0.003	0.019	0.046	0.640

IV. DISCUSSION

The results demonstrated clear tendency of the changes of the EEG rhythms energy between the segments with and without microwave stimulation: these changes became evident (Fig.1) at the EEG rhythm frequencies lower than the modulation frequencies and were statistically significant for some brain regions (Table 1). The results show an increase of the EEG energy level related to the microwave stimulation in all EEG rhythms frequencies (except the alpha rhythm level at 7 Hz frequency modulation). The change caused by the stimulation is higher than natural fluctuations in the reference EEG energy level and increases with modulation frequency (Fig. 1).

In the case of the EEG theta rhythm the effect was more

obvious in frontal and tempora-parietal channels and in the case of the EEG alpha rhythm in occipital channels. The effect was not noticeable at the EEG rhythm frequencies higher than 21 Hz. The experimental data support the idea of frequency dependence of the low-level EMF effect on human brain bioelectrical activity. The mechanisms of this influence are not clear. Processes in synapse, operating as a low-pass filter, can cause the frequency dependence [13]. Similar phenomenon can also be caused by the parametric excitation of the cortical oscillators: in the case of weak parametric factor the excitation effect is optimal in the case of excitation frequency twice higher than the oscillation frequency. The effect was different for different subjects. The reasons may be individual sensitivity and reaction to

stimulation by microwaves. Furthermore, the sensitivity might depend on the variability of the functional state of the brain. The increase of the variability caused by the microwave stimulation has also been reported in our previous work [14]. Clear tendencies of changes in microwave on-off cycles were obvious for some subjects. The same tendency became evident in our previous study on another group of subjects (Fig 7 in [8]).

V. CONCLUSION

The results of the experimental study support the hypothesis raised: the effect of the low-level microwave radiation has a stimulating effect that is dependent of modulation frequency. We can conclude that the modulated low-level microwave radiation causes changes in the EEG rhythms energy at the frequencies lower than the modulation frequency.

The findings allowed us to determine the following tendencies:

- microwave stimulation causes an increase of the EEG energy level;
- the effect is most intense at higher modulation frequencies.

The changes in the EEG rhythm energy level were statistically significant in frontal and tempora-parietal channels for theta and in occipital channels for alpha rhythm frequencies. Individual sensitivity and response to stimulation are very different.

The mechanism of the findings is not clear and the effect needs further investigation.

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